

Appendix A: Obtaining and Installing AI Engine (AIE) License

This appendix provides step-by-step instructions for obtaining, installing, and verifying a node-locked license for the AMD/Xilinx AI Engine (AIE) required for Vitis 2024.

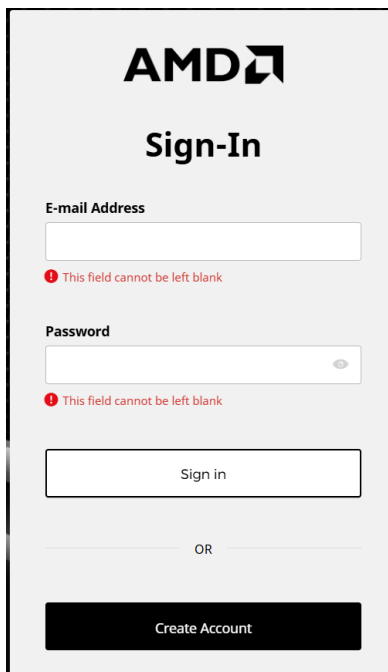
Students must complete this process before running AI Engine simulations.

A.1 Access AMD Licensing Portal

Go to the AMD licensing portal:

<https://account.amd.com/en/forms/license/license-form.html>

Sign in using your AMD/Xilinx account. If you do not have an account, create one and then sign in.

The image shows a screenshot of the AMD Sign-In page. At the top is the AMD logo. Below it is the heading "Sign-In". There are two input fields: "E-mail Address" and "Password". Both fields have a red error message below them that says "This field cannot be left blank". Below the password field is a "Sign in" button. Below the "Sign in" button is the text "OR". At the bottom is a "Create Account" button.

A.2 Create a New License

After signing in, select the “Create New Licenses” tab. You will see a table labeled “Certificate Based Licenses”.

Product Licensing

[Help](#)

Create New Licenses

Manage Licenses



Have a Voucher to Redeem? ?
XXXX-XXXX-XXXX-XXXX

Redeem Now



Evaluation and No Charge Cores ?
Search the **Evaluation** and **No Charge** cores catalog and add specific cores to table below

Search Now

Create a New License File

Create a new license file by making your product selections from the table below. ?

Certificate Based Licenses

	Product	Type	License	Available Seats	Status	Subscription End Date
☐	2025 AI Engine Tools License	Certificate - No Charge	Node	1/1	Current	None
☐	SDSoC Environment, 60 Day Evaluation License	Certificate - Evaluation	Node	1/1	Current	60 days
☐	Vivado ML Enterprise Edition, 30-Day Evaluation License	Certificate - Evaluation	Node	1/1	Current	30 days
☐	Vitis HLS License	Certificate - No Charge	Node	1/1	Current	None
☐	Vitis Model Composer (Xilinx toolbox for MATLAB and Simulink), 90 Day Evaluation	Certificate - Evaluation	Node	1/1	Current	90 days
☐	ISE Embedded Edition License	Certificate - No Charge	Node	1/1	Current	None
☐	SDAccel OpenCL Development Environment: 30 Day Node Locked Evaluation Lice...	Certificate - Evaluation	Node	1/1	Current	30 days
☐	Vivado Design Suite: HL WebPACK 2015 and Earlier License	Certificate - No Charge	Node	1/1	Current	None
☐	ISE WebPACK License	Certificate - No Charge	Node	1/1	Current	None
☐	Xilinx MicroBlaze/All Programmable SoC Software Development Kit – Standalone	Certificate - No Charge	Node	1/1	Current	None
☐	PetaLinux Tools License	Certificate - Evaluation	Node	1/1	Current	365 days

Locate “2025 AI Engine Tools License” and check the box next to it.

Ensure:

- Type = Certificate – No Charge
- License = Node

Scroll down and click “Generate Node-Locked License”.

Create New Licenses

Manage Licenses



Have a Voucher to Redeem? ?
XXXX-XXXX-XXXX-XXXX

Redeem Now



Evaluation and No Charge Cores ?
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☐	Vivado ML Enterprise Edition, 30-Day Evaluation License	Certificate - Evaluation	Node	1/1	Current	30 days
☐	Vitis HLS License	Certificate - No Charge	Node	1/1	Current	None
☐	Vitis Model Composer (Xilinx toolbox for MATLAB and Simulink), 90 Day Evaluation	Certificate - Evaluation	Node	1/1	Current	90 days
☐	ISE Embedded Edition License	Certificate - No Charge	Node	1/1	Current	None
☐	SDAccel OpenCL Development Environment: 30 Day Node Locked Evaluation Lice...	Certificate - Evaluation	Node	1/1	Current	30 days
☐	Vivado Design Suite: HL WebPACK 2015 and Earlier License	Certificate - No Charge	Node	1/1	Current	None
☐	ISE WebPACK License	Certificate - No Charge	Node	1/1	Current	None
☐	Xilinx MicroBlaze/All Programmable SoC Software Development Kit – Standalone	Certificate - No Charge	Node	1/1	Current	None
☐	PetaLinux Tools License	Certificate - Evaluation	Node	1/1	Current	365 days

Generate Node-Locked License

A.3 Enter System Information

After clicking the button:

- You will be prompted to enter:
 - **Host Name**
 - **Host ID (MAC Address (without colons))**

1 PRODUCT SELECTION

Product Selections *	Product	Type	Available Seats	Subscription End Date	Requested Seats	Borrowed Seats
☐	Vitis HLS License	No Charge	1/1	None	1	
☐	SDAccel OpenCL Development Environment: 3...	Evaluation	1/1	30 days	1	
☐	Vivado ML Enterprise Edition, 30-Day Evaluatio...	Evaluation	1/1	30 days	1	
☒	2025 AI Engine Tools License	No Charge	0/1	None	1	
☐	Xilinx MicroBlaze/All Programmable SoC Softw...	No Charge	1/1	None	1	

[Click here](#)

2 SYSTEM INFORMATION

License

Host ID *?

Add a host...

1. Host Name ? :

2. Operating System ? :

3. Host ID Type ? :

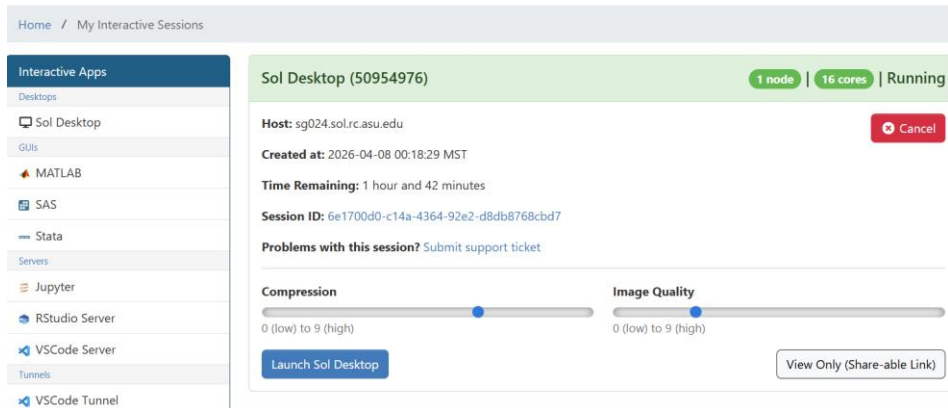
4. Host ID Value ? :

3 COMMENTS

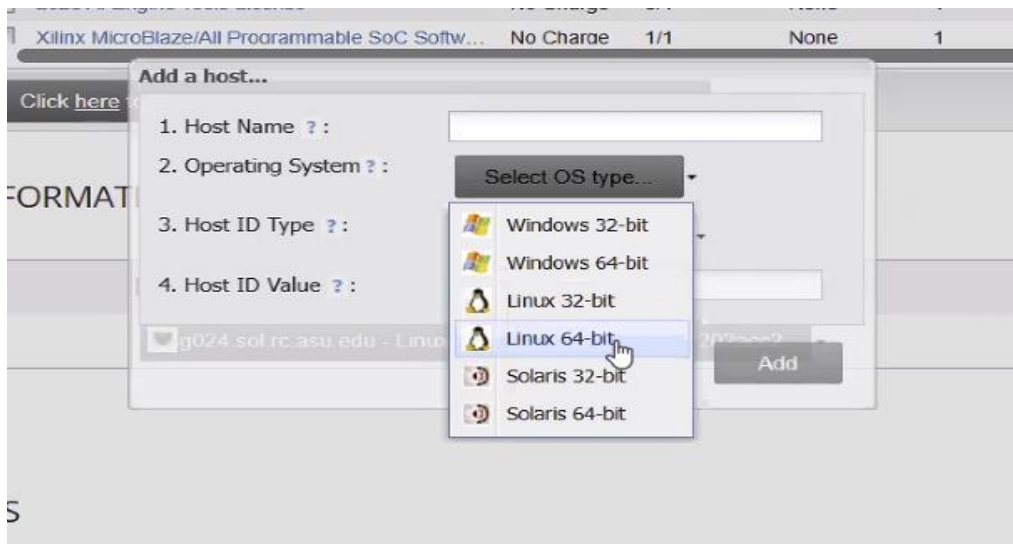
Comments ?

[Cancel](#)

For the Host, you can go to SOL, in the My Interactive Session Window, you can copy and paste the host name. See the figure below.



For the operating system, select Linux 64-bit.



For Host ID Type section select Ethernet MAC.

1

PRODUCT SELECTION

Product Selections *

	Product	Type	Available Seats	Subscription End Date	Requested Seats	Borrowed Seats
☐	ISE WebPACK License	No Charge	1/1	None	1	
☐	PetaLinux Tools License	Evaluation	1/1	365 days	1	
☐	Vivado Design Suite: HL WebPACK 2015 and ...	No Charge	1/1	None	1	
☐	ISE Embedded Edition License	No Charge	1/1	None	1	
☐	Vitis Model Composer (Xilinx toolbox for MATL...	Evaluation	1/1	90 days	1	

Click here

2

SYSTEM INFORMATION

License

Host ID *?

Add a host...

1. Host Name ? :

2. Operating System ? :

Linux 64-bit

3. Host ID Type ? :

Select host type...

4. Host ID Value ? :

Ethernet MAC

g024.sol.rc.asu.edu - Linux 64-bit - Ethernet MAC

Cancel

Add

3

COMMENTS

Comments ?

Launch an SOL node and open a terminal in the lab4 directory. Then run the ifconfig command and press Enter. The first MAC address, which is highlighted in the following figure, should be copied and pasted to the Host ID Value section. Note that you have to remove the : from the address.

```
Terminal - efarahma@sg024:~/Lab4
File Edit View Terminal Tabs Help
(base) [efarahma@sg024:~/Lab4]$ ifconfig
eno8303: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether ec:2a:72:02:ac:c2 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 98

eno8403: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether ec:2a:72:02:ac:c3 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 168

eno12409np1: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether e8:eb:d3:29:4d:61 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

A.4 Download the License

Download the .lic file from the Manage Licenses tab or from your email.

For downloading the license by clicking on the small arrow in the left corner of the manage licenses window, as shown in the following image.

Create New Licenses **Manage Licenses**

Host Name	Host Type	Host ID	License Type	OS	Created By	Created Date
sg024.sol.rc.asu.edu	Ethernet MAC	ec2a7202acc2	Node	Linux 64-bit	EBRAHIM FARAHMA...	08 APR 2026

Page 1 of 1 | Displaying 1 - 1 of 1

Comments

Product	Type	Status	Subscription End Date	Activated Seats
2025 AI Engine Tools License	Certificate - No Charge	Current	None	1

Download

Modify License

A.5 Upload License to SOL

Upload the license file to your home directory on the SOL machine.

A.6 Set Up Environment and Launch Vivado

Then open a terminal in the home directory and use the following command:

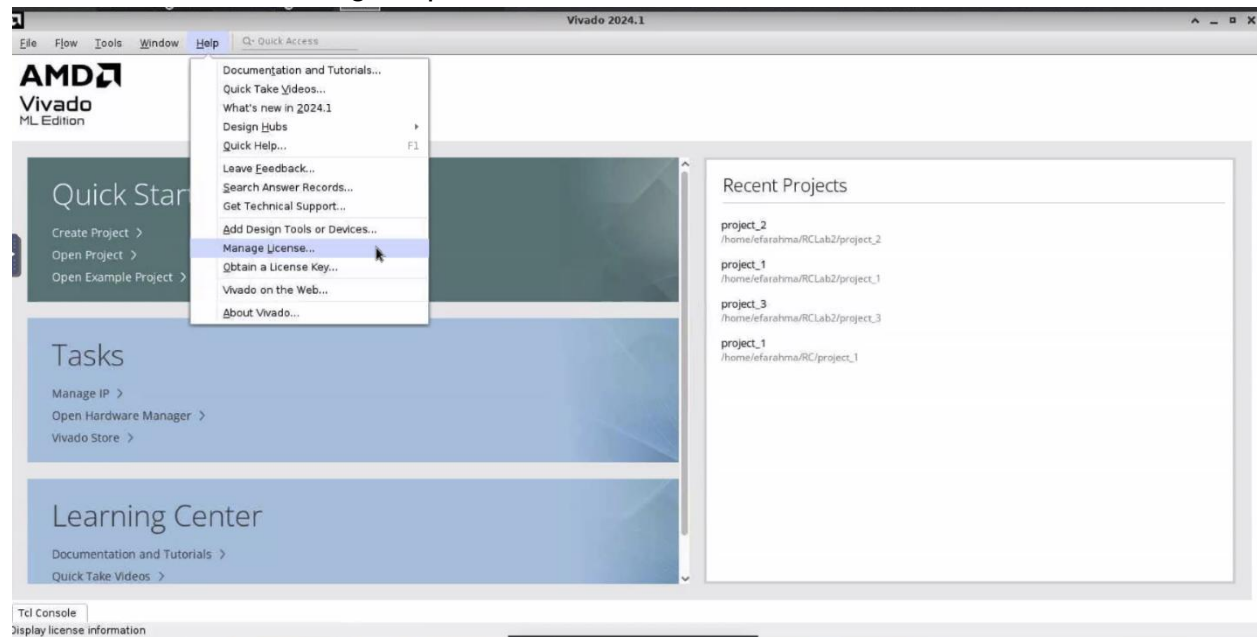
```
source
```

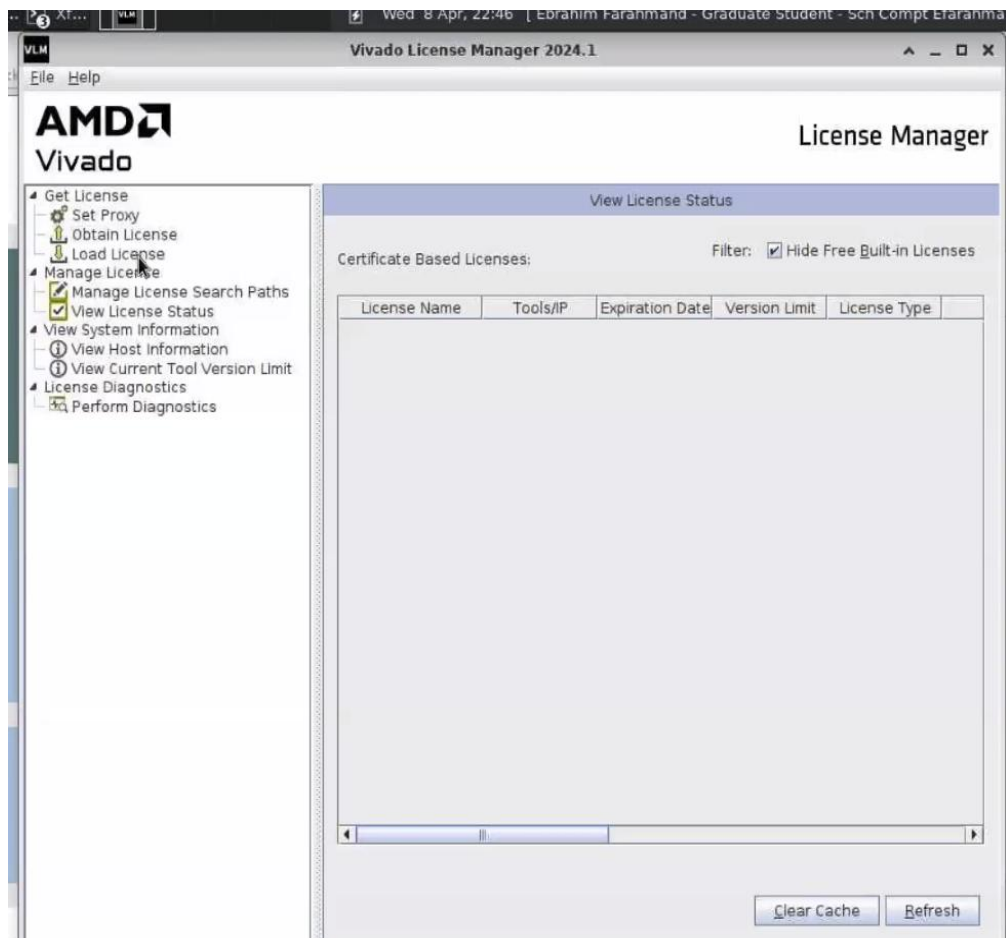
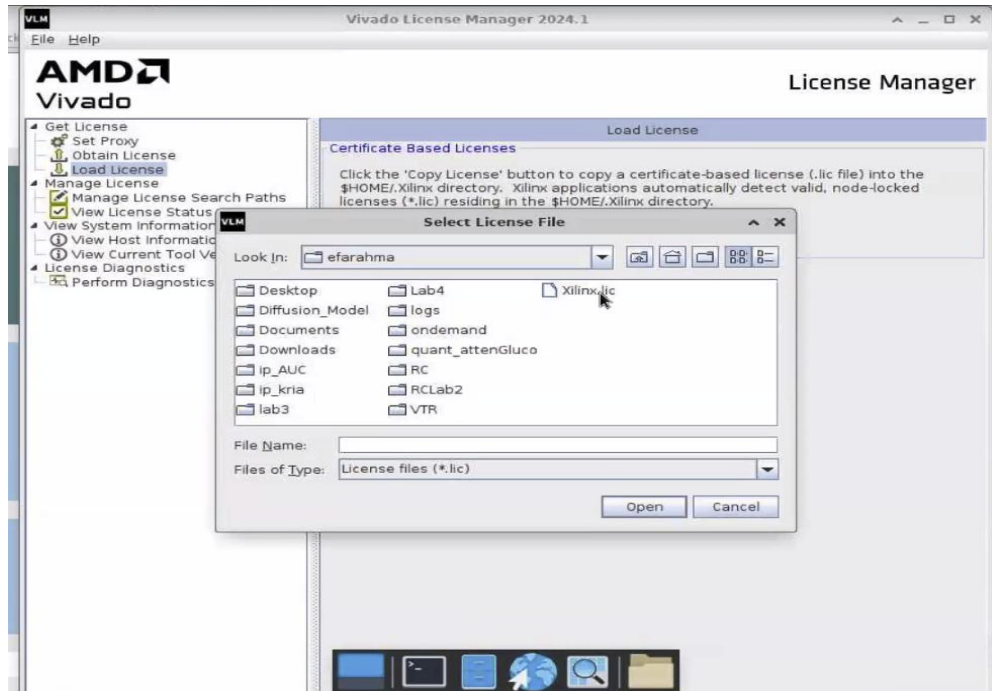
```
/data/courses/class_cse494598cen571spring2026_aaror112/Vitis_2024_1/Vitis/2024.1/settings64.sh
```

Then launch Vivado by using “vivado” command.

A.7 Install License in Vivado

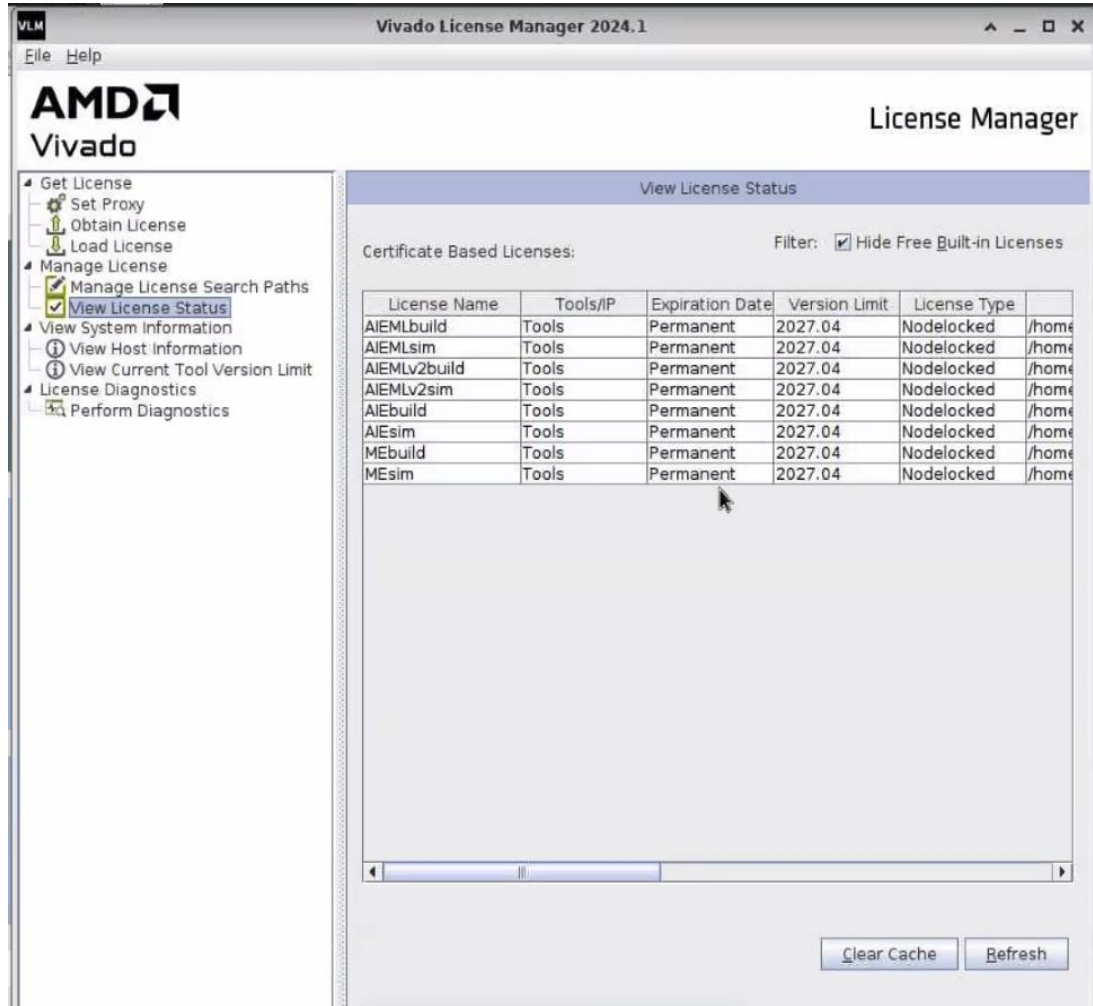
In Vivado, go to Help → Manage License, click Load License, and select your .lic file, as shown in the following snapshots.





A.8 Verify License Installation

Open Vivado and check the license using the License Manager (as shown in the figure). Click on “View License Status” and confirm that the AI Engine license appears in the list with a status of Active/Valid.



A.9 Verify in AIE Simulation

Run the AIE simulator and confirm the license is detected in the output.

FLOW

Component

ale.co...

REPORTS

Run

Debug

REPORTS

AIE SIMULATOR / HARDWARE

Build

REPORTS

Run

Debug

REPORTS

Recent Workspaces

OUTPUT

TASK: AIE SIMULATOR FOR AIE_COMPONENT

PROBLEMS

Terminal will be reused by tasks.

> Executing task: aiesimulator --pkg-dir=./Work '-i ../../' --online -wdb -text --profile --aximm-port=12001 -aie_component/build/hw/aiesimTraceOptions.txt <

Running AIE Simulator in Online mode...

AI Engine Simulator

Version 2024.1 (Linux64-bit)

SW Build 5073938 on 2024-05-20-09:38:44

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AIEMLsim feature license is found.

INFO: Executing config: ./Work/config/sesim_config.json

[INFO]: If you are running trace or profile, Xilinx recommends to use --xlopt=0. This will give a greater vis